QUY NHON SEMICONDUCTORS – QNSC

RTL Design Naming Rule

This document defines mandatory naming rules for RTL design. The rules apply to modules, instances, ports, internal signals, constants, and integration interfaces. Exceptions shall be explicitly reviewed and agreed before implementation.

V1.2

Contents

CONTENTS ..... ERROR! BOOKMARK NOT DEFINED.

1. GENERAL DESIGN PRINCIPLES 6

1.1. Identifier Format 6

1.2. Port Direction Prefix 6

1.3. Active-Low Naming 7

1.4. Numeric Instance and Channel Naming 7

1.5. Standard Vocabulary 7

2. MODULE, INSTANCE, AND INTERNAL RTL NAMING 10

2.1. Module Naming 10

2.2. Instance Naming 11

2.3. Internal Signal Naming 12

2.4. Parameter, Constant, and State Naming 12

2.5. Memory Naming 13

3. PORT AND INTERFACE NAMING 15

3.1. Clock 15

3.2. Reset 15

3.3. APB 15

3.4. AXI 16

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3.4.1. AW Channel 16
3.4.2. W Channel 16
3.4.3. B Channel 16
3.4.4. AR Channel 16
3.4.5. R Channel 17

3.5. CSR 17

3.6. Interrupt 17

3.7. GPIO 18

3.8. Pad 18

3.9. Memory 18

3.10. DMA 19

3.11. JTAG 19

3.12. Debug 19

3.13. Boot 20

3.14. Power 20

3.15. DFT 20

3.16. Analog 21

3.17. Test 21

4. COMPLETE EXAMPLES AND ENFORCEMENT 22

4.1. Complete QNSC Wrapper Example 22

4.2. Complete Naming Example 22

4.3. Enforcement Checklist 23

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Revision

Version	Date	Author	Description
1.0	15/09/2026	Tam Nguyen	Initial version
1.1	28/09/2026	Nghia VT	2.1 and 4.3: a wrapper is m_qnsc_wrap_<ip>, the IP's short name in the chip contract (m_qnsc_wrap_uart, m_qnsc_wrap_isram), not the vendor module name
1.2	01/10/2026	Nghia VT	2.1: chip top m_qnsc_top / m_qnsc_chip, nested wrapper, packages qnsc_<function>_pkg. 2.3: function locals and struct fields. New 2.6 type, 2.7 file, 2.8 technology cell. 3.1/3.2: role as domain in generic modules. 3.3/3.4: buses with several ports, structs inside a block. 4.3: checklist. Examples qnsc_clk_gate, qnsc_rst_sync, u_rst_sync follow 1.5

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1. GENERAL DESIGN PRINCIPLES

1.1. Identifier Format

All RTL identifiers shall**:**

- Use lowercase English letters.
- Use digits where required.
- Use underscore _ as the only word separator.
- Not use CamelCase, PascalCase, or mixed-case naming.
- Use meaningful and consistent terminology throughout the project.

Examples

Valid	Invalid
`m_qnsc_timer`	`M_QNSC_TIMER`
`i_bus_apb_paddr`	`iApbAddr`
`o_int_timer_0`	`o_irq_timer0`
`i_rst_n_sys`	`rstn_sys`
`u_timer_0`	`uTimer0`

1.2. Port Direction Prefix

All module ports shall use the following prefixes:

Prefix	Meaning	Example
`i_`	Input	`i_clk_sys`
`o_`	Output	`o_int_timer_0`
`io_`	Bidirectional	`io_i2c_sda`

The direction prefix shall always be placed at the beginning of the port name.

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1.3. Active-Low Naming

Active-low signals shall contain `_n` immediately after the signal meaning.

Examples:

```
i_rst_n_sys
i_rst_n_cpu
o_rst_n_peri
```

`_n` shall not be replaced by inconsistent forms such as:

```
rstn
resetn
reset_b
rst_bar
```

1.4. Numeric Instance and Channel Naming

Numeric instances or channels shall use an underscore before the index.

Valid

```
timer_0
timer_1
uart_0
uart_1
dma_0
dma_1
```

Invalid

```
timer0
timer1
uart0
dma0
```

1.5. Standard Vocabulary

Project-wide standard vocabulary shall be used consistently.

Function	Mandatory terminology

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Function	Mandatory terminology
Interrupt	`int`
Power	`pwr`
Design-for-Test	`dft`
Analog	`ana`
Multiplexer	`mux`
Clock	`clk`
Reset	`rst`
Configuration	`cfg`
Debug	`dbg`
Test	`test`
Boot	`boot`
Memory	`mem`
Peripheral	`peri`
UART	`uart`
SPI	`spi`
I2C	`i2c`
GPIO	`gpio`
DMA	`dma`

Standard protocol vocabulary shall be preserved, including:

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```
paddr
pwdata
prdata
pready
psel
penable
pwrite

awaddr
awvalid
awready
wdata
wstrb
wvalid
wready
bresp
bvalid
bready
araddr
arvalid
arready
rdata
rresp
rvalid
rready
```

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2. MODULE, INSTANCE, AND INTERNAL RTL NAMING

2.1. Module Naming

QNSC functional modules shall use:

```
m_qnsc_<function>
```

Examples:

```
m_qnsc_timer
m_qnsc_dma
m_qnsc_gpio
m_qnsc_uart
```

QNSC wrapper modules shall use:

```
m_qnsc_wrap_<ip>
```

where `<ip>` is the short name of the IP as the chip contract names it, without an index, not the name of the vendor module. The name stays the same when the IP is replaced. Two configurations of one IP are two IPs in the contract, each with its own wrapper.

Examples:

```
m_qnsc_wrap_cpu
m_qnsc_wrap_uart
m_qnsc_wrap_timer
m_qnsc_wrap_isram
m_qnsc_wrap_dsram
```

Generic modules shall use:

```
qnsc_<function>
```

Examples:

```
qnsc_fifo_sync
qnsc_arbiter
qnsc_clk_gate
qnsc_rst_sync
```

The chip top shall use:

```
m_qnsc_top
m_qnsc_chip
```

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m_qnsc_top instantiates every block wrapper; m_qnsc_chip is m_qnsc_top plus the pad ring. Both are defined in design/top/rtl/ and nowhere else. The top of a block is its wrapper, m_qnsc_wrap_<ip>; no _top suffix is added.

A wrapper nested inside a block wrapper, around one IP of a block built from several, shall use:

```
m_qnsc_wrap_<ip>_<part>
```

Examples:

```
m_qnsc_wrap_cpu_ibex
m_qnsc_wrap_cpu_cpu2axi
```

Packages shall use:

```
qnsc_<function>_pkg
```

Examples:

```
qnsc_pkg
qnsc_cpu2axi_pkg
```

Packages share one namespace with every vendored package (axi_pkg, tlul_pkg); the prefix keeps ours apart. qnsc_pkg is the package generated from the chip contract.

2.2. Instance Naming

Module instances shall use:

```
u_<function>_<index>
```

Examples:

```
u_timer_0
u_timer_1
u_uart_0
u_dma_0
u_fifo_0
```

For a single non-indexed instance:

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```
u_timer
u_cdc
u_rst_sync
```

2.3. Internal Signal Naming

Internal registered signals shall use:

```
r_<function>
```

Internal combinational signals shall use:

```
w_<function>
```

Examples:

```
r_timer_count
r_timer_enable
r_state

w_timer_done
w_timer_match
w_next_state
```

Example

```
reg  [31:0] r_timer_count;
wire        w_timer_done;
```

A variable local to a function or a task, and a field of a struct, is not an internal signal and takes no prefix.

2.4. Parameter, Constant, and State Naming

Parameters shall use:

```
P_<FUNCTION>
```

Examples:

```
P_DATA_WIDTH
P_ADDR_WIDTH
P_DEPTH
```

Local constants shall use:

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```
C_<FUNCTION>
```

Examples:

```
C_IDLE
C_BUSY
C_ERROR
```

FSM states shall use:

```
S_<STATE>
```

Examples:

```
S_IDLE
S_WAIT
S_WRITE
S_READ
```

Example:

```
localparam P_DATA_WIDTH = 32;

localparam C_IDLE  = 2'd0;
localparam C_BUSY  = 2'd1;
```

2.5. Memory Naming

Memory arrays shall use:

```
mem_<function>
```

Examples:

```
mem_data
mem_tag
mem_fifo
mem_instruction
```

2.6. Type Naming

Types shall use:

```
<function>_t
```

Examples:

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```
state_t
addr_t
axi_req_t
```

The members of an enum are constants or FSM states and follow 2.4:

```
typedef enum logic [1:0] {S_IDLE, S_BUSY, S_DONE} state_t;
```

2.7. File Naming

Each file shall hold one module, package or interface, and shall be named after it:

```
m_qnsc_wrap_uart.sv     module m_qnsc_wrap_uart
qnsc_cpu2axi_pkg.sv     package qnsc_cpu2axi_pkg
```

The source of a wrapper generated with emacs is <module>.src.sv. Simulation, synthesis and connectivity flows take the top of a block from the file name, so a file named otherwise selects the wrong top.

2.8. Technology Cell Naming

RTL shall not instantiate a cell of a technology library (standard cell, I/O pad, memory macro) directly. It instantiates a generic module, which each technology implements with the same name and ports:

```
qnsc_<function>
```

Examples:

```
qnsc_clk_gate
qnsc_clk_buf
qnsc_clk_inv
```

Inside an implementation, the library cell keeps its own name and pins, and its instance shall use:

```
u_size_only_<function>
```

so that synthesis can preserve it by name. How a technology is added and chosen: design/common/tech/README.md.

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3. PORT AND INTERFACE NAMING

3.1. Clock

Clock ports shall use:

```
i_clk_<domain>
o_clk_<domain>
```

Examples:

```
i_clk_sys
i_clk_cpu
i_clk_peri
o_clk_peri
```

In a generic module (qnsc_<function>), which belongs to no domain, <domain> names the role of the clock: i_clk_src, i_clk_dst, o_clk_gated. A port named i_clk alone is not allowed.

3.2. Reset

Reset ports shall use:

```
i_rst_n_<domain>
o_rst_n_<domain>
```

Examples:

```
i_rst_n_sys
i_rst_n_cpu
i_rst_n_peri
o_rst_n_peri
```

Every reset port is active low. In a generic module, <domain> names the role, as for clocks: i_rst_n_dst.

3.3. APB

APB ports shall use:

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```
i_bus_apb_<signal>
o_bus_apb_<signal>
```

Examples:

```
i_bus_apb_paddr
i_bus_apb_psel
i_bus_apb_penable
i_bus_apb_pwrite
i_bus_apb_pwdata

o_bus_apb_prdata
o_bus_apb_pready
o_bus_apb_pslverr
```

A module with several APB ports names the port after the protocol, as the chip contract names it (APB_M8 is m_8):

```
o_bus_apb_m_8_psel
i_bus_apb_m_8_prdata
```

3.4. AXI

AXI naming shall be normalized by individual channel and signal**.**

The format shall be:

```
i_bus_axi_<channel>_<signal>
o_bus_axi_<channel>_<signal>
```

3.4.1. AW Channel

```
i_bus_axi_aw_addr
i_bus_axi_aw_valid
o_bus_axi_aw_ready
```

3.4.2. W Channel

```
i_bus_axi_w_data
i_bus_axi_w_strb
i_bus_axi_w_valid
o_bus_axi_w_ready
```

3.4.3. B Channel

```
o_bus_axi_b_resp
o_bus_axi_b_valid
i_bus_axi_b_ready
```

3.4.4. AR Channel

```
i_bus_axi_ar_addr
i_bus_axi_ar_valid
```

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```
o_bus_axi_ar_ready
```

3.4.5. R Channel

```
o_bus_axi_r_data
o_bus_axi_r_resp
o_bus_axi_r_valid
i_bus_axi_r_ready
```

A module with several AXI ports names the port after the protocol, as the chip contract names it (AXI_S0 is s_0, AXI_M1 is m_1):

```
i_bus_axi_s_0_aw_addr
o_bus_axi_m_1_aw_addr
```

A module inside a block may carry a bus as one packed struct per direction: o_bus_axi_req, i_bus_axi_rsp (and the same for APB). The ports of a wrapper stay one per signal, as above.

3.5. CSR

CSR ports shall use:

```
i_csr_<function>
o_csr_<function>
```

Examples:

```
i_csr_timer_enable
i_csr_timer_prescaler

o_csr_timer_status
o_csr_timer_count
```

3.6. Interrupt

Interrupt ports shall use:

```
i_int_<source>
o_int_<source>
```

Examples:

```
o_int_timer_0
```

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```
o_int_uart_0
o_int_dma_0

i_int_external
i_int_software
i_int_timer
```

3.7. GPIO

GPIO ports shall use:

```
i_gpio_<function>
o_gpio_<function>
io_gpio_<function>
```

Examples:

```
i_gpio_uart_rx
o_gpio_spi_sclk
o_gpio_i2c_sda
```

3.8. Pad

Pad ports shall use:

```
i_pad_<function>
o_pad_<function>
io_pad_<function>
```

Examples:

```
i_pad_usi_0
o_pad_can_sel
io_pad_i2c_scl
```

3.9. Memory

Memory ports shall use:

```
i_mem_<function>
o_mem_<function>
```

Examples:

```
o_mem_addr
```

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```
o_mem_wdata
o_mem_we
o_mem_be
i_mem_rdata
```

3.10. DMA

DMA ports shall use:

```
i_dma_<function>
o_dma_<function>
```

Examples:

```
o_dma_req
i_dma_gnt
i_dma_rvalid
o_dma_addr
o_dma_wdata
i_dma_rdata
```

3.11. JTAG

JTAG ports shall use:

```
i_jtag_<function>
o_jtag_<function>
```

Examples:

```
i_jtag_tck
i_jtag_tms
i_jtag_tdi
o_jtag_tdo
```

3.12. Debug

Debug ports shall use:

```
i_dbg_<function>
o_dbg_<function>
```

Examples:

```
i_dbg_req
```

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```
i_dbg_enable
o_dbg_ack
o_dbg_halted
```

3.13. Boot

Boot ports shall use:

```
i_boot_<function>
o_boot_<function>
```

Examples:

```
i_boot_addr
i_boot_sel
o_boot_done
```

3.14. Power

Power-related ports shall use:

```
i_pwr_<function>
o_pwr_<function>
```

Examples:

```
i_pwr_sleep
i_pwr_isolation
o_pwr_good
```

3.15. DFT

DFT ports shall use:

```
i_dft_<function>
o_dft_<function>
```

Examples:

```
i_dft_scan_en
i_dft_bypass
o_dft_scan_out
```

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3.16. Analog

Analog ports shall use:

```
i_ana_<function>
o_ana_<function>
io_ana_<function>
```

Examples:

```
i_ana_comp_out
o_ana_trim
io_ana_test
```

3.17. Test

Test-related ports shall use:

```
i_test_<function>
o_test_<function>
```

Examples:

```
i_test_en
i_test_mode
o_test_status
```

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4. COMPLETE EXAMPLES AND ENFORCEMENT

4.1. Complete QNSC Wrapper Example

```
module m_qnsc_wrap_timer (
    input  wire        i_clk_sys,
    input  wire        i_rst_n_sys,

    input  wire [15:0] i_bus_apb_paddr,
    input  wire        i_bus_apb_psel,
    input  wire        i_bus_apb_penable,
    input  wire        i_bus_apb_pwrite,
    input  wire [31:0] i_bus_apb_pwdata,

    output wire [31:0] o_bus_apb_prdata,
    output wire        o_bus_apb_pready,
    output wire        o_bus_apb_pslverr,

    output wire        o_int_timer_0
);
```

Internal RTL:

```
reg  [31:0] r_timer_count;
reg         r_timer_enable;

wire        w_timer_done;
wire        w_timer_match;

localparam P_DATA_WIDTH = 32;

localparam C_IDLE = 2'd0;
localparam C_BUSY = 2'd1;

reg [1:0] r_state;
```

Instance:

```
timer u_timer_0 (
    ...
);
```

4.2. Complete Naming Example

A timer module integrated through APB shall follow:

```
Module
    m_qnsc_wrap_timer

Clock
    i_clk_sys

Reset
```

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```
    i_rst_n_sys

APB
    i_bus_apb_paddr
    i_bus_apb_psel
    i_bus_apb_penable
    i_bus_apb_pwrite
    i_bus_apb_pwdata
    o_bus_apb_prdata
    o_bus_apb_pready
    o_bus_apb_pslverr

Interrupt
    o_int_timer_0

Internal registered signal
    r_timer_count

Internal combinational signal
    w_timer_done

Parameter
    P_DATA_WIDTH

Constant
    C_IDLE

Instance
    u_timer_0
```

4.3. Enforcement Checklist

Before RTL is committed, the following rules shall be satisfied:

Check	Mandatory requirement
Identifier case	All identifiers shall be lowercase
Word separator	Words shall use _
Port direction	Ports shall use i_, o_, or io_, matching the direction
QNSC module	Shall use `m_qnsc_<function>`
QNSC wrapper	Shall use `m_qnsc_wrap_<ip>`
Instance	Shall use `u_<function>_<index>` where indexed

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Check	Mandatory requirement
Registered signal	Shall use `r_<function>`
Combinational signal	Shall use `w_<function>`
Parameter	Shall use `P_<FUNCTION>`
Constant	Shall use `C_<FUNCTION>`
FSM state	Shall use `S_<STATE>`
Numeric instance	Shall use `timer_0`, not `timer0`
AXI	Shall be normalized by channel and signal
Pad	Shall use `i_pad_`, `o_pad_`, `io_pad_`
Power	Shall use `pwr`
DFT	Shall use `dft`
Analog	Shall use `ana`
Mux	Shall use `mux`
Third-party IP	Shall be normalized at wrapper boundary
Chip top	Shall be m_qnsc_top or m_qnsc_chip, in design/top/rtl/
Package	Shall use qnsc_<function>_pkg
Type	Shall use <function>_t; enum members S_<STATE> or C_<FUNCTION>
File	One module, package or interface per file, named after it

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Check	Mandatory requirement
Clock and reset	Shall use i_clk_<domain> and i_rst_n_<domain>
Bus with several ports	Shall use <i|o>_bus_<protocol>_<port>_..., <port> as in the contract
Technology cell	Shall be instantiated through qnsc_<function>; the library cell only inside, as u_size_only_<function>

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